Python and Git 101

Python Programming and Data Analysis · Day 1 · Dataminded Academy

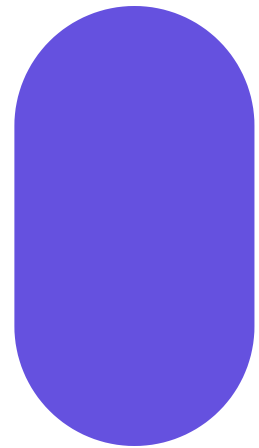
dataminded

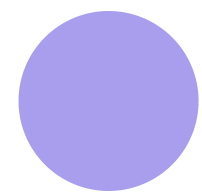
NICE TO MEET YOU

Jan Vanbuel

Your trainer today. Ask early, ask often: stuck for five minutes means wave at me.

Dataminded





Today covers three things you will use every week



Development environment

Python, uv, VS Code, PowerShell



Programming fundamentals

Types, loops, functions, files



Version control

Git and Azure DevOps

> No programming experience assumed. Nobody is behind.

AI writes code, you ask, judge and choose

Ask

You have to ask in the specialist's **vocabulary**

Judge

You have to read what comes back and decide if it is **right**

Choose

There are several approaches, and someone must own the **tradeoff**

> "Loop over the CSV files in this folder and return a dict per run" beats "make it work for all my files".

Use Copilot, keep only lines you can explain

Explain errors

Paste the error and ask what it means and where to look

Ask for a second way

Then compare the two and pick the simpler one

Own the commit

Your name is on it. Explain every line to your neighbour, or replace it

Judge code in this order: correct, readable, fast



1 · Correct

Does it do what it should?



2 · Readable

Will you understand it in two months?



3 · Fast

Is it fast enough? Ask this last.

By 17:00 you can do six things on your own laptop

Run a script

Set up Python and run a script from PowerShell

Manage a project

A virtual environment per project, packages with uv

Write a program

Functions, conditions, loops and imports, run from VS Code

Read an error

Find the line a stack trace points to

Automate files

Read a folder of files and write a summary

Work with Git

Clone, commit on a branch, push and pull from Azure DevOps



Agenda

01	09:00 Setup check
02	09:30 Scripts, notebooks and running Python
03	10:15 Language fundamentals, lunch at 12:30
04	13:15 Standard library, files and exceptions
05	14:20 Projects with uv
06	15:05 Version control and Git
07	16:50 Wrap-up



Four commands prove your laptop is ready

```
uv --version
uv python list --only-installed
code --version
git --version
```

uv
Python and packages

Python 3.13
installed by uv

VS Code
the editor

Git
version control

Something fails?
Do not debug alone. Wave at me: setup problems are normal on managed laptops.

Scripts and notebooks



A script is a text file that runs top to bottom

```
# bench_summary.py
pressure_bar = 7.5

power_kw = 37.5
print(f"Power: {power_kw} kW at {pressure_bar} bar")

print("Done")
```

```
Power: 37.5 kW at 7.5 bar
Done
```

Scripts repeat, notebooks explore

	Script (.py)	Notebook (.ipynb)
Runs	Start to finish, every time, on demand or scheduled	Cell by cell, in any order you click
Great for	Automation, reusable tools, anything others run	Exploring data, plotting, explaining results
Weak at	Slow feedback when exploring	Repeatability and testing
In Git	Clean text, easy to compare versions	JSON with embedded output, noisy differences



A notebook can run code you no longer see

```
In [1]   limit_bar = 9.0
```

```
In [2]   limit_bar = 6.0
```

ran, then deleted

```
In [3]   print(limit_bar)
```

```
Out      6.0
```

not 9.0

- You ran cell 1, then cell 2, then edited cell 2 away and forgot about it
- Cell 3 prints a value that no longer appears anywhere in the visible code
- A colleague runs the notebook top to bottom and gets a different answer

Words you can now use script · notebook · cell · kernel · hidden state · reproducible

Python on your laptop

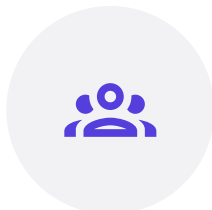


Python is open source and runs on CPython



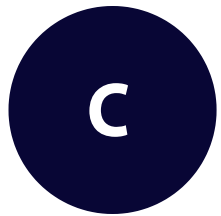
1991

by Guido van Rossum, named after Monty Python



Open source

steered by the PSF



CPython 3.13

the interpreter, written in C

Your .py file → **Compiled to bytecode** → **CPython runs it**

Check which Python a command really runs

```
Get-Command python
```

```
CommandType  Name        Source
-----------  ----        ------
Application  python.exe  C:\Users\you\AppData\Local\Microsoft\WindowsApps\python.exe
```

Watch out

A `WindowsApps` path is the Microsoft Store shortcut, not a real Python. It opens the Store or does nothing useful.

Go to the folder, then uv run the script

```
# hello.py
print("Compressor test bench online")
```

```
cd C:\Users\you\training
uv run python hello.py
```

```
Compressor test bench online
```

VS Code runs the same script in three steps

1

OPEN THE FOLDER

File, Open Folder, choose your project folder

2

SELECT THE INTERPRETER

Command Palette, Python: Select Interpreter, pick the one in `.venv` or the uv Python 3.13

3

RUN

Play button at the top right, or `uv run python hello.py` in the integrated terminal



Exercise 01: First script

`exercises/python/01-first-script`

Words you can now use interpreter · CPython · terminal · PATH · working folder · uv run

Language fundamentals

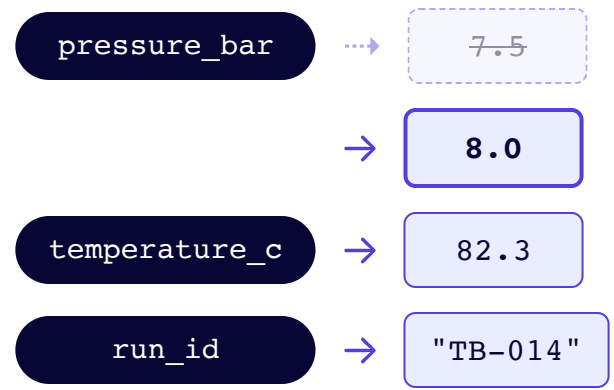


Variables are names for values

```
pressure_bar = 7.5
temperature_c = 82.3
run_id = "TB-014"

pressure_bar = 8.0
print(pressure_bar)
print(temperature_c)
```

```
8.0
82.3
```



Every value has a type

```
run_id = "TB-014"        # str: text
cycles = 12              # int: whole number
flow_l_s = 18.4          # float: decimal number
is_stable = True         # bool: True or False
operator = None          # None: no value (yet)

print(type(flow_l_s))
```

```
<class 'float'>
```

Text that looks like a number is still text

```
text = "7.5"
pressure_bar = float(text)
print(pressure_bar + 1)        # 8.5

count = 3
print("Cycles: " + count)
```

```
TypeError: can only concatenate str (not "int") to str
```




f-strings put values inside text

```
run_id = "TB-014"
pressure_bar = 7.5
print(f"Run {run_id}: {pressure_bar:.1f} bar")
print(f"{pressure_bar * 14.5038:.0f} psi")
print("TB" in run_id, run_id.lower())
```

```
Run TB-014: 7.5 bar
109 psi
True tb-014
```




Put the unit in the name

Written fast

```
p = 7.5
t = 82
calc(p, t)
```

Read easily

```
pressure_bar = 7.5
temperature_c = 82
to_kelvin(temperature_c)
```

Put the unit in the name. Constants in capitals: `ALARM_LIMIT_BAR = 8.5`



Exercise 02: Variables and types

`exercises/python/02-variables-and-types`

Words you can now use variable · assignment · type · string · float · f-string · convert

A list keeps values in order

```
pressures_bar = [6.8, 7.1, 7.5, 7.4, 7.9]
print(pressures_bar[0])
print(len(pressures_bar))
pressures_bar.append(8.2)
print(max(pressures_bar))
```

```
6.8
5
8.2
```




Indexing starts at zero

index	0	1	2	3	4
value	6.8	7.1	7.5	7.4	7.9
negative	-5	-4	-3	-2	-1

```
p = [6.8, 7.1, 7.5, 7.4, 7.9]
print(p[1:3], p[:2], p[-2:])
```

```
[7.1, 7.5] [6.8, 7.1] [7.4, 7.9]
```




A dict finds values by name

```
reading = {"run": "TB-014", "pressure_bar": 7.5}
print(reading["pressure_bar"])
reading["flow_l_s"] = 18.4
print(reading.get("power_kw"))
print(reading)
```

```
7.5
None
{'run': 'TB-014', 'pressure_bar': 7.5, 'flow_l_s': 18.4}
```




A list for a series, a dict for a record

	list	dict
Find by	Position, in order	Name, the key
Fits	A series of the same kind of value	One record with different fields
Example	`pressures_bar[2]`	`reading["pressure_bar"]`

> Copilot will pick one for you unless you say which you want.



Exercise 03: Collections

`exercises/python/03-collections`

Words you can now use list · index · slice · dict · key · value



if runs the first branch that is true

```
pressure_bar = 9.4
if pressure_bar > 9.0:
    print("ALARM: over limit")
elif pressure_bar < 5.0:
    print("Warning: low pressure")
else:
    print("OK")
```

```
ALARM: over limit
```

`pressure_bar > 9.0`

true: run this branch, skip the rest

`pressure_bar < 5.0`

not checked

`else`

not checked

Checked top to bottom, the first true one wins



for runs the block once per element

```
pressures_bar = [6.8, 7.1, 9.4]
for p in pressures_bar:
    print(f"{p:.1f} bar")

for i in range(3):
    print(i)
```

```
6.8 bar
7.1 bar
9.4 bar
0
1
2
```




Most scripts initialise, loop, test, collect

```
pressures_bar = [6.8, 7.1, 9.4, 7.4, 9.1]
high = []
total = 0.0

for p in pressures_bar:
    total += p
    if p > 9.0:
        high.append(p)

print(high, round(total / len(pressures_bar), 2))
```

```
[9.4, 9.1] 7.96
```

1. **INITIALISE**
   An empty list and a zero total
2. **LOOP**
   Take each value in turn
3. **TEST**
   Decide with an `if`
4. **COLLECT**
   Report at the end



Exercise 04: Control flow

`exercises/python/04-control-flow`

Words you can now use condition · branch · loop · iterate · block · indentation



A function names a piece of logic

```
def to_psi(pressure_bar):
    return round(pressure_bar * 14.5038, 1)

print(to_psi(7.5))
print(to_psi(10))
```

```
108.8
145.0
```



`return` passes a value on, `print` only shows it

```
def double_and_print(x):
    print(x * 2)

result = double_and_print(4)
print(result)
```

```
8
None
```

A function without `return` gives back `None`. Use `print` to show, `return` to pass on.

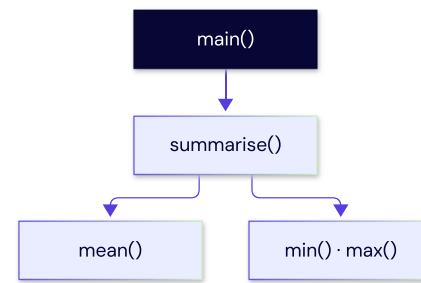
A script is small functions plus a main

```
def mean(values):
    """Return the arithmetic mean of a list of numbers."""
    return sum(values) / len(values)

def summarise(values):
    return {"min": min(values), "max": max(values), "mean": mean(values)}

def main():
    pressures_bar = [6.8, 7.1, 9.4, 7.4]
    print(summarise(pressures_bar))

if __name__ == "__main__":
    main()
```



Type hints tell you how to call a function

```python
def to_psi(pressure_bar: float, decimals: int = 1) -> float:
    return round(pressure_bar * 14.5038, decimals)
```

pressure_bar: float

The parameter and the type it expects

decimals: int = 1

A type and a default value, so you may leave it out

-> float

The type of the value that comes back

Hints are documentation that the editor and tools can check. Python does not enforce them when the script runs.

Move code outward only when you need to

	Use it when	The cost
Lines in the script	You do it once, in a few lines	Copy and paste when you need it again
A function	You repeat it, or it deserves a name	A little structure to read through
A module	Several scripts need the same functions	One more file, and an `import`

> Copilot will pick one for you unless you say which you want.



A test says what correct means

```python
import pytest
import conversions

def test_bar_to_psi():
    assert conversions.bar_to_psi(1) == pytest.approx(14.5038)
```

Red: run, read the failure → Green: make it pass → Test first: write the next test

> This loop is **test-driven development**.

Exercise 05: Functions

`exercises/python/05-functions`

Words you can now use function · argument · return value · type hint · test · assertion · test-driven development



The standard library



import gives you ready-made tools like math

```
import math

diameter_m = 0.05
area_m2 = math.pi * (diameter_m / 2) ** 2
flow_m3_s = 18.4 / 1000
print(round(flow_m3_s / area_m2, 2))

print(math.isclose(0.1 + 0.2, 0.3))
```

```
9.37
True
```




pathlib builds paths and finds files

```
from pathlib import Path

data_dir = Path("data")
log_file = data_dir / "run_014.csv"

print(log_file.name, log_file.suffix, log_file.stem)
print(log_file.exists())

for f in sorted(data_dir.glob("*.csv")):
    print(f)
```

```
run_014.csv .csv run_014
True
data\run_014.csv
data\run_015.csv
```




csv turns each row into a dict of strings

```
t_s,pressure_bar,temperature_c
0,6.8,71.2
10,7.1,74.5
20,7.5,78.9
```

```
import csv
with open("data/run_014.csv", newline="", encoding="utf-8") as f:
    rows = list(csv.DictReader(f))
print(rows[0])
print(float(rows[1]["pressure_bar"]) + 1)
```

```
{'t_s': '0', 'pressure_bar': '6.8', 'temperature_c': '71.2'}
8.1
```




json stores settings and nested data

```
import json
config = {"limit_bar": 9.0, "channels": ["pressure", "flow"], "operator": None}
text = json.dumps(config)
print(text)
back = json.loads(text)
print(back["channels"][0], back["operator"])
```

```
{"limit_bar": 9.0, "channels": ["pressure", "flow"], "operator": null}
pressure None
```




Exercise 06: Standard library

`exercises/python/06-standard-library`

Words you can now use standard library · module · import · path · glob · CSV · JSON



Files and exceptions



with open reads a file and closes it again

```
from pathlib import Path

with open("data/run_014.csv", encoding="utf-8") as f:
    for line in f:
        print(line.strip())

text = Path("data/run_014.csv").read_text(encoding="utf-8")
```

```
t_s,pressure_bar,temperature_c
0,6.8,71.2
10,7.1,74.5
20,7.5,78.9
```




Mode w overwrites, mode a appends

```
lines = ["run,max_pressure_bar", "TB-014,7.9"]
with open("summary.txt", "w", encoding="utf-8") as f:
    for line in lines:
        f.write(line + "\n")

with open("summary.txt", "a", encoding="utf-8") as f:
    f.write("TB-015,8.2\n")
```



Mode `"w"` replaces an existing file without asking. `"a"` adds at the end, `"r"` only reads.

A traceback is a map to the error

```
Traceback (most recent call last):
  File "convert.py", line 2, in <module>
    pressure_bar = float(text)
ValueError: could not convert string to float: 'seven'
```

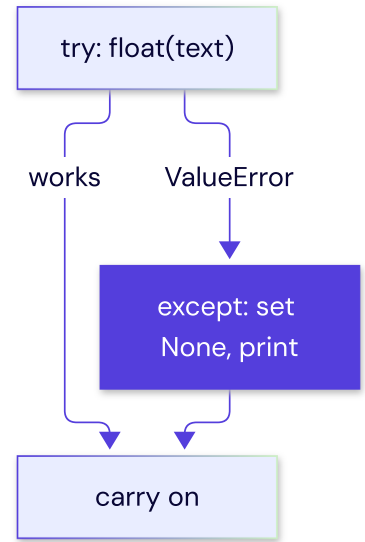
- Line 1 is only a header, it says: read on
- The middle shows the file, the line number and the code that failed
- The last line names the error type and what went wrong. Read it first.



try / except handles an error you expect

```
text = "seven"
try:
    pressure_bar = float(text)
except ValueError:
    pressure_bar = None
    print(f"Skipping bad value: {text!r}")
print(pressure_bar)
```

```
Skipping bad value: 'seven'
None
```



Exercise 07: Files and exceptions

`exercises/python/07-files-and-exceptions`

Words you can now use file handle · encoding · exception · traceback · raise · catch

Projects with uv



uv and pip: the same job, fewer steps

	pip workflow	uv workflow
Environment	`python -m venv .venv`	`uv init`, then automatic
Activate	`.venv\Scripts\Activate.ps1`	Not needed: `uv run`
Install	`pip install matplotlib`	`uv add matplotlib`
Declare	`requirements.txt`, by hand	`pyproject.toml`, updated by uv
Pin versions	`pip freeze`, if you remember	`uv.lock`, automatic



Two commands give a project its own packages

```
uv init bench-tools
cd bench-tools
uv add matplotlib
uv run python main.py
```

```
Creating virtual environment at: .venv
Resolved 12 packages in 330ms
Installed 11 packages in 68ms
 + matplotlib==3.11.2
 + numpy==2.5.4
 ...
```




pyproject.toml declares, uv.lock pins

```toml
[project]
name = "bench-tools"
requires-python = ">=3.13"
dependencies = [
    "matplotlib>=3.11.2",
]
```

Commit

`pyproject.toml`, `uv.lock`, `.python-version` and your `.py` files

Do not commit

`.venv/`, big and machine specific



Exercise 08: A uv project

`exercises/python/08-uv-project`

Words you can now use virtual environment · package · dependency · lockfile · pin · sync



Python goes further, but not today

- **Styles**: procedural (today), object-oriented with classes and dataclasses, functional
- **Language**: decorators, generators and comprehensions, context managers
- **Debugging**: the VS Code debugger, with breakpoints and stepping, instead of `print`
- **Scripts that grow**: `logging` instead of `print`, arguments with `argparse`

- **Performance**: measure first (`timeit`, a profiler), then NumPy
- **Concurrency**: threads, processes, `asyncio`
- **Extending**: C, or Rust through PyO3
- **Packaging**: your own code, installable by colleagues



Version control



final_v3 is version control by hand

```
\\fileserver\compressors\testbench\
    analyse_run.py
    analyse_run_final.py
    analyse_run_final_v3_USE_THIS_ONE.py
```

With AI: `git diff` shows exactly what an assistant changed, so you keep what is good and undo the rest.

History

Which version made last week's report?

Who and why

Who changed the conversion factor?

Undo

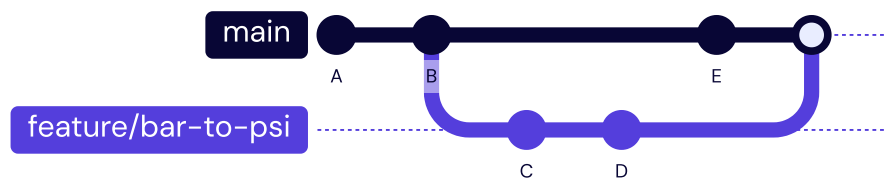
How do I get back to what worked?

Collaborate

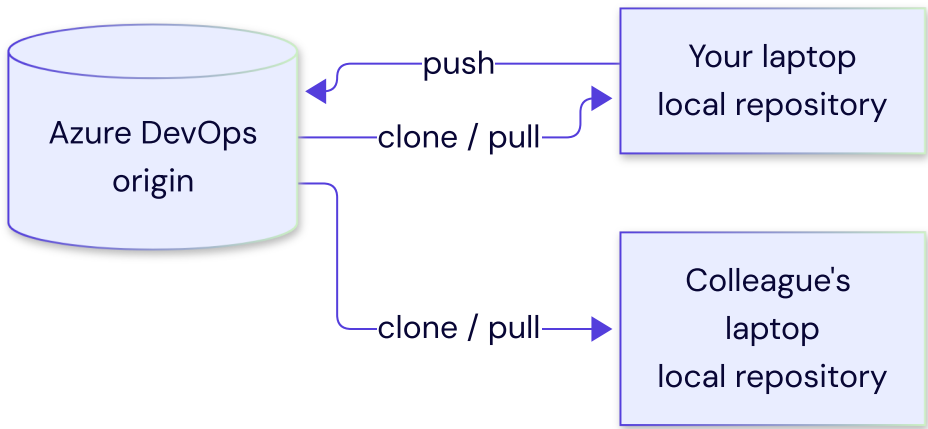
Two people edit, whose version wins?

A branch is a movable label on a commit

- **Repository**: a project folder plus its full history
- **Commit**: a saved snapshot with a message, author and date
- **Branch**: a movable label on a commit
- Work on a branch does not touch `main` until you merge it

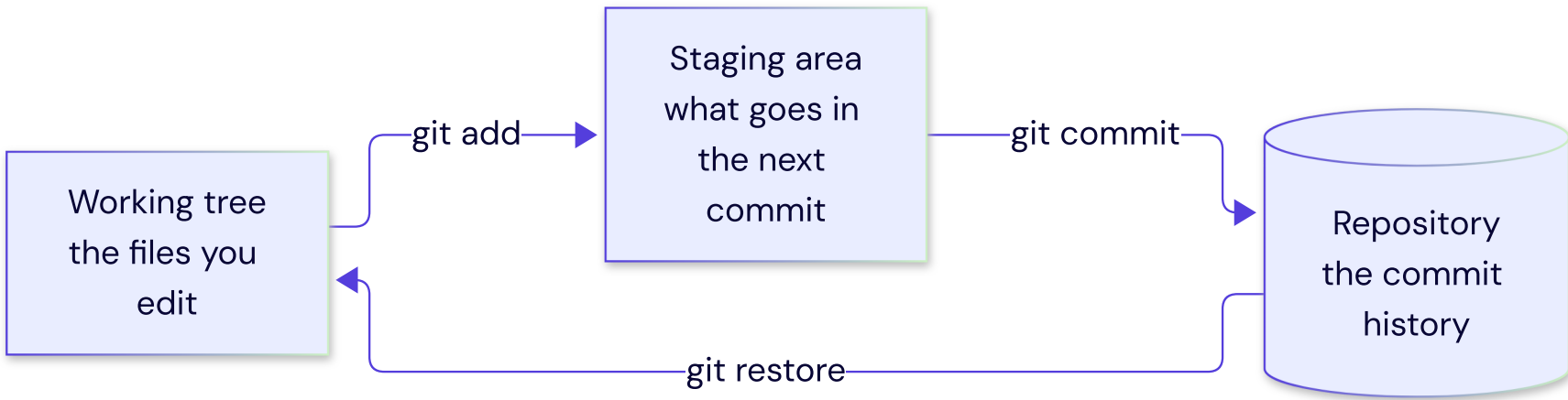


Azure DevOps hosts the shared remote



- **Git** is the free tool on your laptop, **Azure DevOps** hosts the shared copy
- A **remote** is that shared copy, by convention called `origin`

A change moves through three areas



- You choose what goes into a commit by **staging** it first
- `git status` tells you which area each change is in

Words you can now use repository · commit · branch · remote · working tree · staging area

Git on the command line



Set your identity once, then init a repository

```
git config --global user.name "Your Name"
git config --global user.email "you@atlascopco.com"
mkdir testbench
cd testbench
git init -b main
git status
```

- Identity: once per laptop, same email as your Azure DevOps account
- `git status` never changes anything: run it all the time

Stage what belongs together, then commit

```
git add conversions.py
git commit -m "Add psi to bar conversion"
```

```
[main (root-commit) a055887] Add psi to bar conversion
 1 file changed, 2 insertions(+)
```

- Message: short, imperative, one logical change: "Fix psi to bar factor", not "update"
- `git add .` stages everything: check `git status` first



log and diff show what changed

```
git log --oneline
```

```
9b221ad Add first measurement
a055887 Add psi to bar conversion
```

```
git diff            # edits not yet staged
git diff --staged   # what the next commit will contain
```

- If the output is long, Git opens a pager: **q** leaves it



Exercise time: first commit

`exercises/git/01-first-commit`

About 10 minutes: `uv run python exercises/git/setup_exercise.py 1`



show and restore bring back old versions

```
git show a055887                    # one commit: message and diff
git show HEAD~1:conversions.py      # a file as it was one commit ago
git restore --source=HEAD~1 conversions.py
git restore conversions.py          # discard edits since last commit
```

- `HEAD` is the latest commit, `HEAD~1` the one before it
- `git restore <file>` throws your edits away: there is no undo for that



Exercise time: history

`exercises/git/02-history`

About 10 minutes: `uv run python exercises/git/setup_exercise.py 2`



Work on a branch, then merge it into main

```
git switch -c feature/bar-to-psi
git add conversions.py
git commit -m "Add bar to psi conversion"
git switch main
git merge feature/bar-to-psi
```

```
Updating 9b221ad..1cd9279
Fast-forward
```

- **Fast-forward** if `main` did not move; otherwise Git makes a merge commit
- `git branch` lists branches, `git branch -d <name>` deletes one



A conflict means Git needs you to decide

```
CONFLICT (content): Merge conflict in conversions.py

def psi_to_bar(psi):
<<<<<<< HEAD
    return psi * 0.068948
=======
    return psi * 0.06895
>>>>>>> feature/a
```

- A conflict is Git asking you to decide, nothing is lost
- Edit the file, delete the marker lines, then `git add` and `git commit`
- Changed your mind? `git merge --abort`



Exercise time: branches

`exercises/git/03-branches`

About 15 minutes: `uv run python exercises/git/setup_exercise.py 3`

Words you can now use stage · commit message · diff · HEAD · merge · fast-forward · conflict



Remotes and Azure DevOps



Clone copies the repository and its history

```
git clone https://dev.azure.com/<organisation>/<project>/_git/<training-repo>
cd <training-repo>
git remote -v
```

```
origin  https://dev.azure.com/<organisation>/<project>/_git/<training-repo> (fetch)
```

- The first time, a browser window asks you to sign in (Git Credential Manager)
- `clone` creates the remote `origin`; the URL is on the **Clone** button in Azure DevOps



Push uploads commits, pull downloads them

```
git push -u origin feature/bar-to-psi   # first push of a new branch
git push                                 # later pushes
git pull                                 # fetch and merge new commits
git fetch                                # only download, change nothing
```

- `-u` links your local branch to the remote one, needed only once
- `pull` is `fetch` plus merge: commit or restore your edits first



Exercise time: Azure DevOps remote

`exercises/git/04-azure-devops-remote`

About 15 minutes: clone, branch, commit, push



A pull request lets a colleague review first



Propose

A pull request (PR) proposes merging your branch into `main`. Colleagues read the diff first.

Complete

Pick the merge type (squash keeps `main` readable) and tick **Delete source branch**.

Update your laptop

`git switch main`, then `git pull`

Exercise time: pull request

`exercises/git/05-pull-request`

About 20 minutes: PR into main, review, complete, pull



.gitignore keeps files out of history

```
# .gitignore
.venv/
__pycache__/
data/raw/
```

- One pattern per line; commit the file itself so everyone shares it
- Never commit passwords, tokens or large raw data
- Day 2 uses it for data files and notebooks



Three tools drive the same Git

	Command line	VS Code	SourceTree
Advantages	Every feature, same everywhere, easy to copy into notes	Already open, diff and conflict views built in	Clear graph of branches, point and click
Disadvantages	Commands to learn, easy to mistype	Hides the commands, fewer rare features	Extra install, less common on managed laptops
Use it when	You need full power, or help from a colleague or a doc	Daily edit, commit, push work next to the code	You want to browse a complex history visually

Demo: Git in VS Code

Optional practice: `exercises/git/06-git-in-vscode`

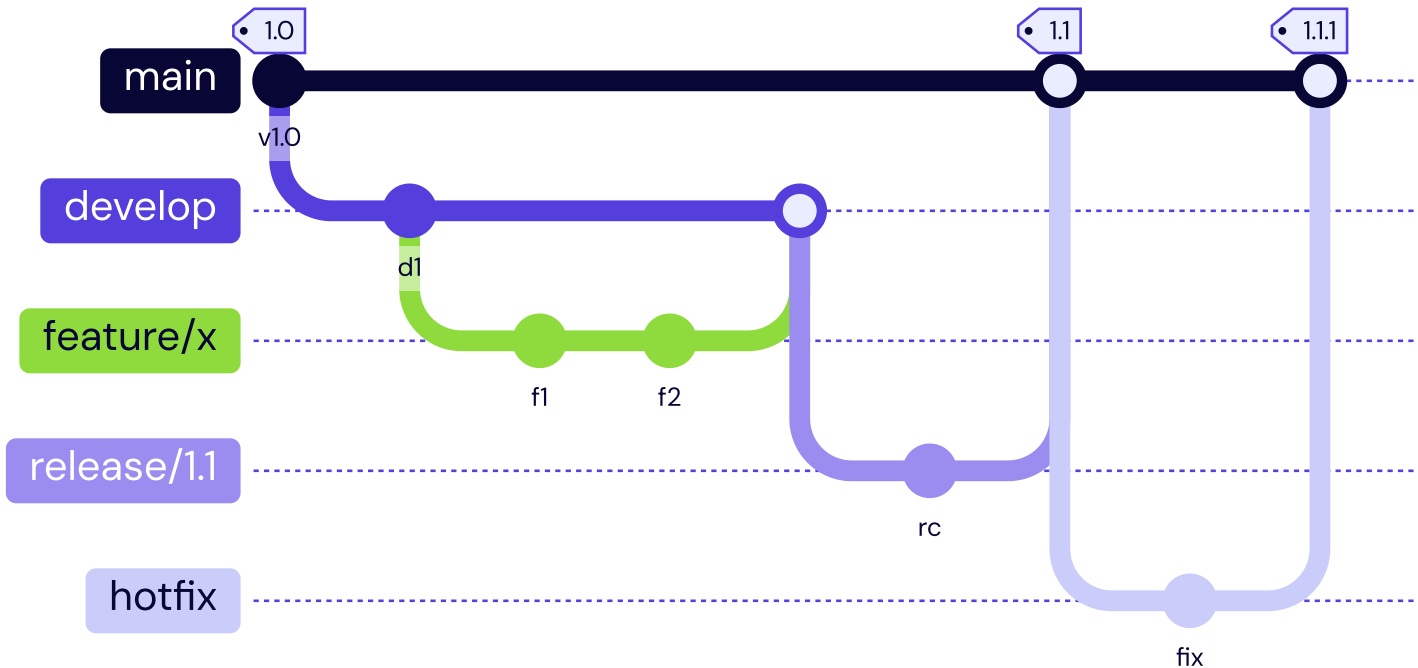
Words you can now use clone · origin · push · pull · fetch · pull request · review



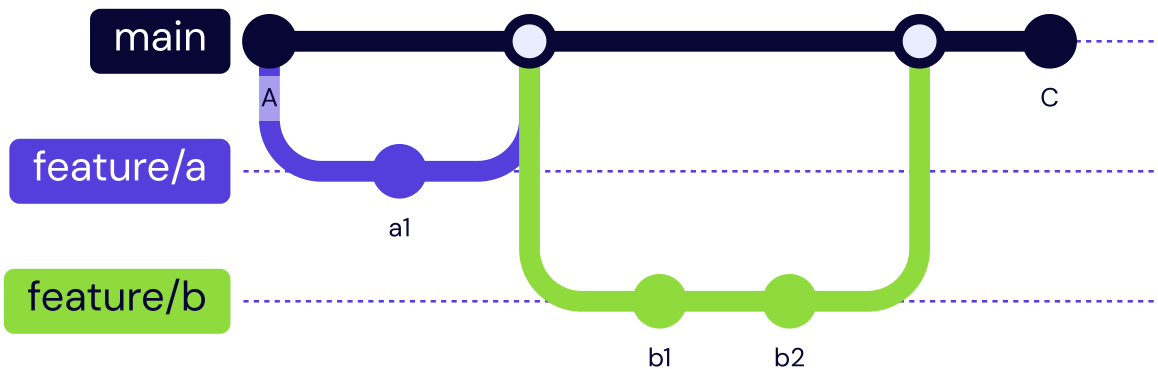
Branching models



GitFlow keeps a branch per release stage



Trunk-based: one main, short branches



- One long-lived branch, `main`; short branches merged through a pull request

Use trunk-based for scripts and analyses

	GitFlow	Trunk-based
Fits	Versioned products, several releases in support	Continuous delivery, scripts, analyses, small teams
Cost	Many branches, large merges, more conflicts	Discipline: small changes, review, tests

- **Use trunk-based**: short-lived `feature/...` branches and PRs; day 3 adds a branch policy on `main`
- Exists, not today: `git rebase`, `git cherry-pick`, `git stash`

Words you can now use trunk-based · GitFlow · feature branch · short-lived · branch policy

You can now do all six things

Run a script

From PowerShell and VS Code

Manage a project

A .venv per project with uv

Write a program

Functions, loops, imports

Read an error

Find the line in a stack trace

Automate files

Read files, write a summary

Work with Git

Branch, commit, push, PR, pull

Today: the **vocabulary** and the tradeoffs. Day 3: you use them to direct and review Copilot and coding agents.

Eight commands cover daily Git

Local

```
git status
git add <file>
git commit -m "message"
git log --oneline
git diff
git restore <file>
git switch -c <branch>
git merge <branch>
```

Remote

```
git clone <url>
git switch main
git pull
git switch -c feature/x
git push -u origin feature/x
# open the PR, complete it, repeat
```

Before day 2: one full cycle with a PR

Before day 2

- A working uv environment
- Redo the fundamentals you found hard
- One full clone, commit, push, pull cycle with a PR

Read more

- Pro Git book (`git-scm.com/book`)
- Learn Git Branching (`learngitbranching.js.org`)
- Python tutorial (`docs.python.org/3/tutorial`)
- uv docs (`docs.astral.sh/uv`)

Next days

- Day 2: pandas, NumPy, SciPy, Matplotlib, scikit-learn
- Day 3, going deeper: tests, linting, pipelines, coding agents and reviewing their work





Thank you

See you on day 2

